Algorithmic Pseudocode Sample 3

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

for $t \in \{-1, \cdots, -T^{\text{traceback}}\}$ **do** ▷ Retrieve the values of $P_{u,t}$

  $P_{u,t} \leftarrow \texttt{Power.GetValue}(t)$

end for

for $t \in \{-1, \cdots, -T^{\text{traceback}}\}$ **do** ▷ Initial conditions on the state variables

  if $P_{u,t} > \texttt{MinimumPower.GetValue}(t)$ **then**

    $S_{u,t}^{\text{OFF}} \leftarrow 0$

    $S_{u,t}^{\text{STOP}} \leftarrow 0$

    $S_{u,t}^{\text{START}} \leftarrow 0$

    if $P_{u,t} < P_{u,t-1}$ **then** ▷ Exact initialization required only if the FLAT state is defined.

      $S_{t-1}^{\text{ON_UP}} \leftarrow 0$

      $S_{t-1}^{\text{ON_DOWN}} \leftarrow 1$

      $S_{t-1}^{\text{ON_FLAT}} \leftarrow 0$

    else if $P_{u,t} > P_{u,t-1}$ **then**

      $S_{t-1}^{\text{ON_UP}} \leftarrow 1$

      $S_{t-1}^{\text{ON_DOWN}} \leftarrow 0$

      $S_{t-1}^{\text{ON_FLAT}} \leftarrow 0$

    else if $P_{u,t} = P_{u,t-1}$ **then**

      $S_{t-1}^{\text{ON_UP}} \leftarrow 0$

      $S_{t-1}^{\text{ON_DOWN}} \leftarrow 0$

      $S_{t-1}^{\text{ON_FLAT}} \leftarrow 1$

    end if

else if $P_{u,t} > 0$ **then** ▷ Reconstruct the startups and shutdowns

if $P_{u,t} < P_{u,t-1}$ **then**

$S_t^{\text{STOP}} \leftarrow 1$

$S_t^{\text{START}} \leftarrow 0$

else

$S_t^{\text{STOP}} \leftarrow 0$

$S_t^{\text{START}} \leftarrow 1$

end if

else ▷ Final possibility: the unit is OFF.

$S_{u,t}^{\text{OFF}} \leftarrow 1$

$S_{u,t}^{\text{STOP}} \leftarrow 0$

$S_{u,t}^{\text{START}} \leftarrow 0$

$S_{t-1}^{\text{ON_UP}} \leftarrow 0$

$S_{t-1}^{\text{ON_DOWN}} \leftarrow 0$

$S_{t-1}^{\text{ON_FLAT}} \leftarrow 0$

end if

end for

Algorithm 2 Initialization of the control and state variables

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